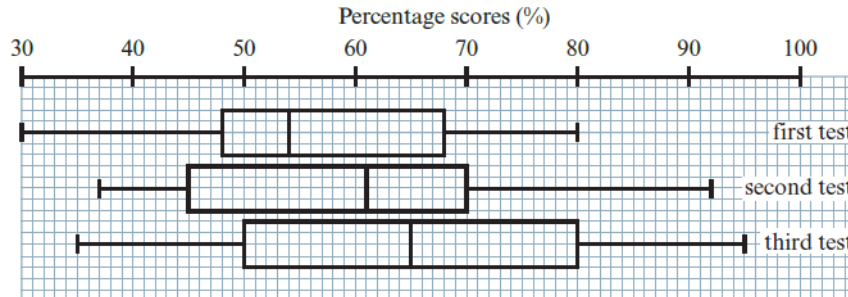




United International University
School of Science and Engineering
Department of Computer Science and Engineering
 Mid Term Examination || Trimester: Summer – 2025
 Course: Math -2205 (Probability & Statistics)
 Total marks – 30 || Duration – 1 hour 30 minutes

[Note that the number of marks is given in brackets [] at the end of each question or part question. You are requested to answer all the questions in order.]

- Q1 (a)** The following box plots summarize the percentage scores of a class of students in the three Mathematics tests they took this term



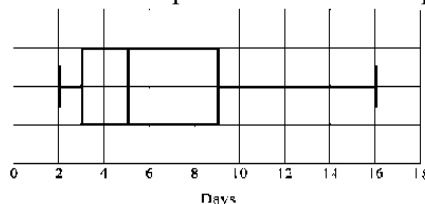
- (i) Find the median and interquartile range of the first test score. [2]
 (ii) How much higher is the upper (third) quartile of the third test compared to the first test? [1]
 (iii) Showing necessary working state, the nature of the distribution of the second test scores? [2]
- (b) The first four central moments of distribution are 0, 7, 26 and 39 respectively. Calculate the standard deviation and skewness and hence comment on the nature of the distribution. [2]
- (c) Survey of a random sample of 20 people recorded the number of unwanted emails they received in a particular week. The results are given below. [3]
- | | | | | | | | | | |
|----|----|----|----|----|----|----|----|----|----|
| 18 | 13 | 18 | 21 | 17 | 22 | 27 | 8 | 11 | 26 |
| 32 | 17 | 31 | 20 | 36 | 15 | 13 | 25 | 35 | 29 |
- Represent the data in a stem and leaf diagram and hence find the median.

- Q2 (a)**

Height h (cm)	$0 \leq h < 10$	$10 \leq h < 20$	$20 \leq h < 30$	$30 \leq h < 40$	$40 \leq h \leq 50$
Frequency	3	12	9	6	5

Using the data given in the above table

- (i) Identify the median class and find the percentage whose heights are below 20 cm. [2]
 (ii) Calculate the mode of the distribution. [3]
- (b) A health worker plotted a box-and whisker plot of the times that patients stayed in hospital.



Based on the diagram, determine the possible range of times that could be considered as outliers. [2]

P.T.O.

(c) Two shopkeepers sell rice.

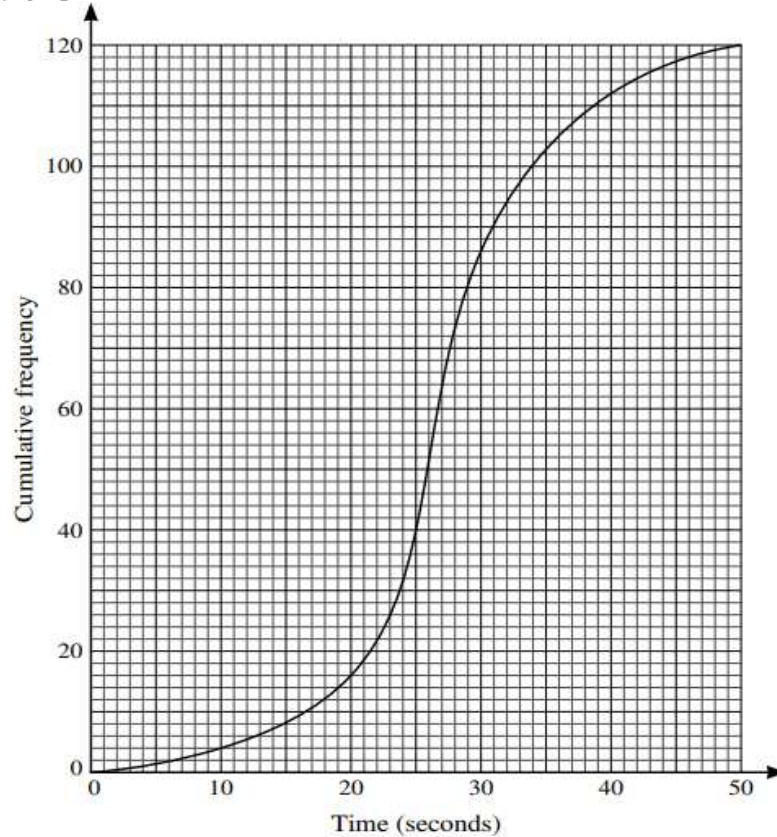
Shopkeeper A records the daily sales (in kg) over 5 days as: 48, 52, 50, 49, 51.

[5]

Shopkeeper B records the mean daily sales over the 5 days of is 50 kg and standard deviation is 10 kg.

Calculate the **Coefficient of Variation (CV)** for both shopkeepers and determine whose sales are more consistent.

- Q3 (a)** The times taken by 120 children to complete a particular puzzle are represented in the following cumulative frequency graph.



- (i) Use the graph to estimate how many of the students took between 20 to 30 seconds. [1]

- (ii) 30% of the children took more than T seconds to complete the puzzle. Use the graph to estimate the value of T . [2]

- (b) A company wants to analyze the relationship between the number of hours studied (x) and the score obtained in a test (y) by 6 students. The data collected is as follows:

Hour Studies (x)	2	3	5	8	9	10
Test Score (y)	65	70	75	85	95	98

- (i) Using the method of least squares, find the equation of the linear regression line in the form: [4]

$$y = a + bx$$

- (ii) Using the above equation, determine the number of study hours required to achieve a score of 90. [1]

Formulae:	For the regression line: $y = a + bx$	$a = \frac{\sum y \sum x^2 - \sum x \sum xy}{n \sum x^2 - (\sum x)^2}$ and $b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$
	Coefficient of correlation	$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$
	Spearman's rank correlation coefficient:	$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$